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SIMATS ENGINEERING
ASSESSMENT TOOL 2: Concept Mapping

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1521

COURSE OUTCOME COVERED

CO1: Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 4

Duration: 60 Minutes

Total Marks: 20

Code of Conduct

192365046

I Pranav Reddy (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Pranav

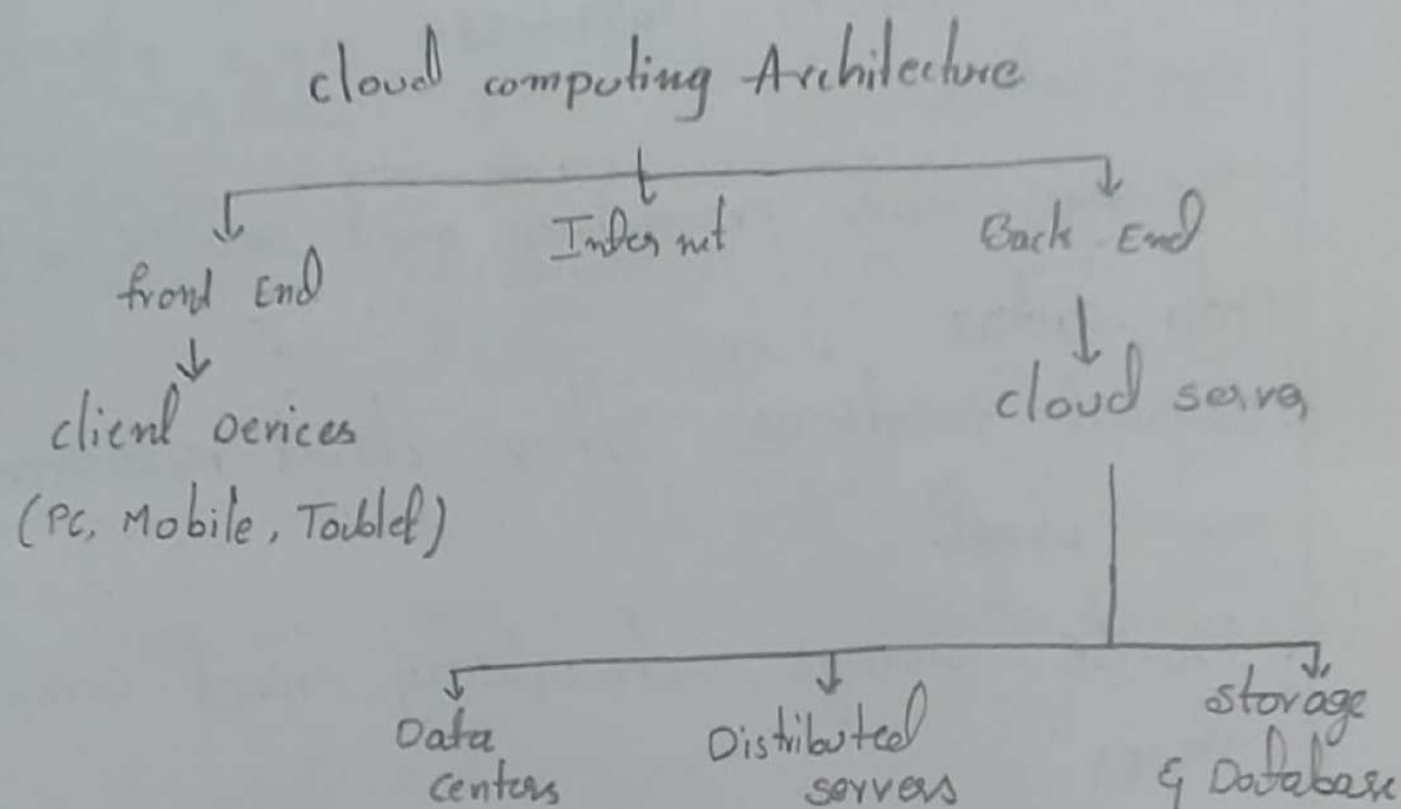
Criteria	Concept Identification (1.25)	Relationship Mapping (1.25)	Hierarchy and Organization (1)	Technical Relevance (1)	Visual Clarity (0.5)	TOTAL (5)
Weightage	25%	25%	20%	20%	10%	100%
Q1						
Q2						
Q3						
Q4						

Signature of the Faculty

Total Marks Awarded:

cloud computing Architecture: Include front-end client devices, data centers and distributed servers.

cloud computing Architecture.



Front End:

- It is the user interface used to access cloud servers
- Includes web browsers and mobile applications.

client Devices:

Devices such as laptops, smart phones, tablets

and PCs.

- Used by users to connect to the cloud through internet.

Back-End

- Contains servers, databases and cloud applications.
- Manages data processing and service delivery

Data centers

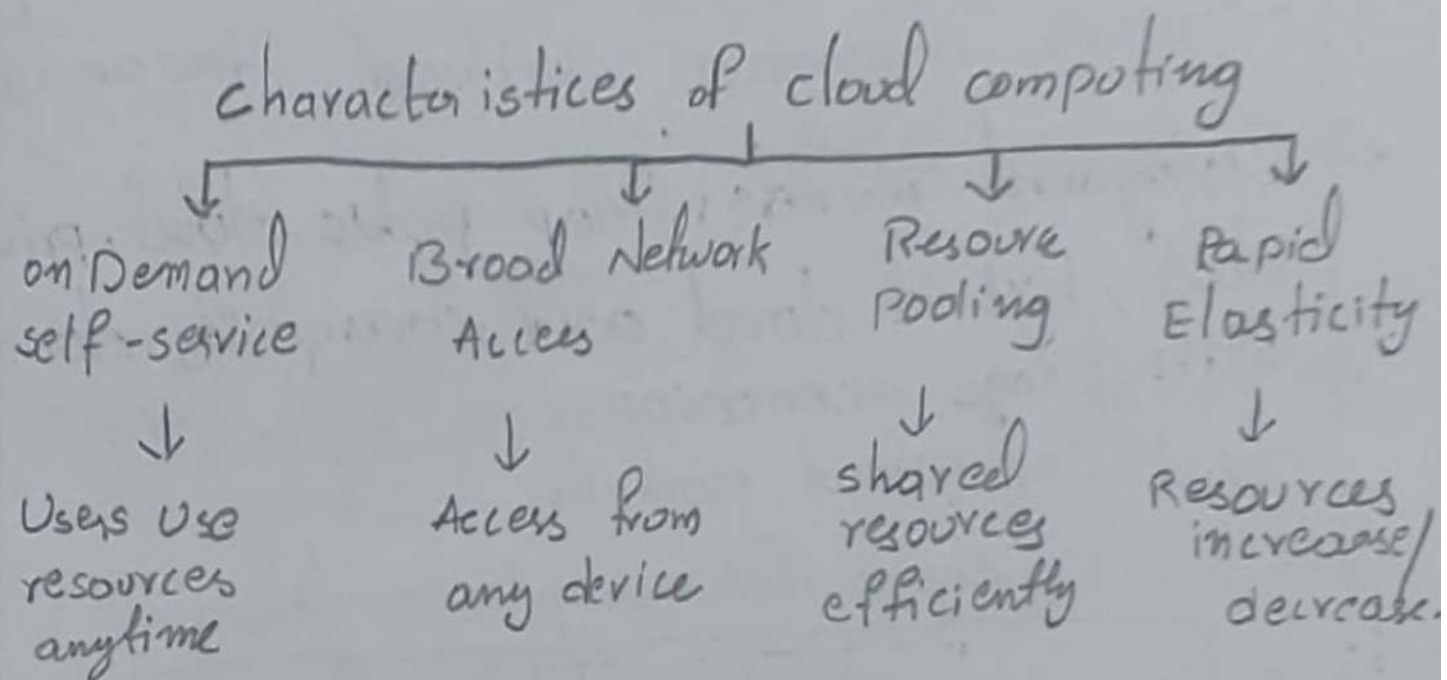
- physical locations where cloud resources are stored.
- Provide storage, networking, and computing power

Distributed Servers:

- Servers locations where cloud resources are stored.
- Improve speed, reliability and availability of services.

characteristics of cloud computing: on-demand self service, broad network access, resource pooling, rapid elasticity and measured service.

characteristics of cloud computing:



On Demand self service:

- Users can access cloud resources when needed.
- No human interaction with the provider is required.

Broad Network Access:

- services are available through the internet
- can be accessed using different devices

Resource Poding:

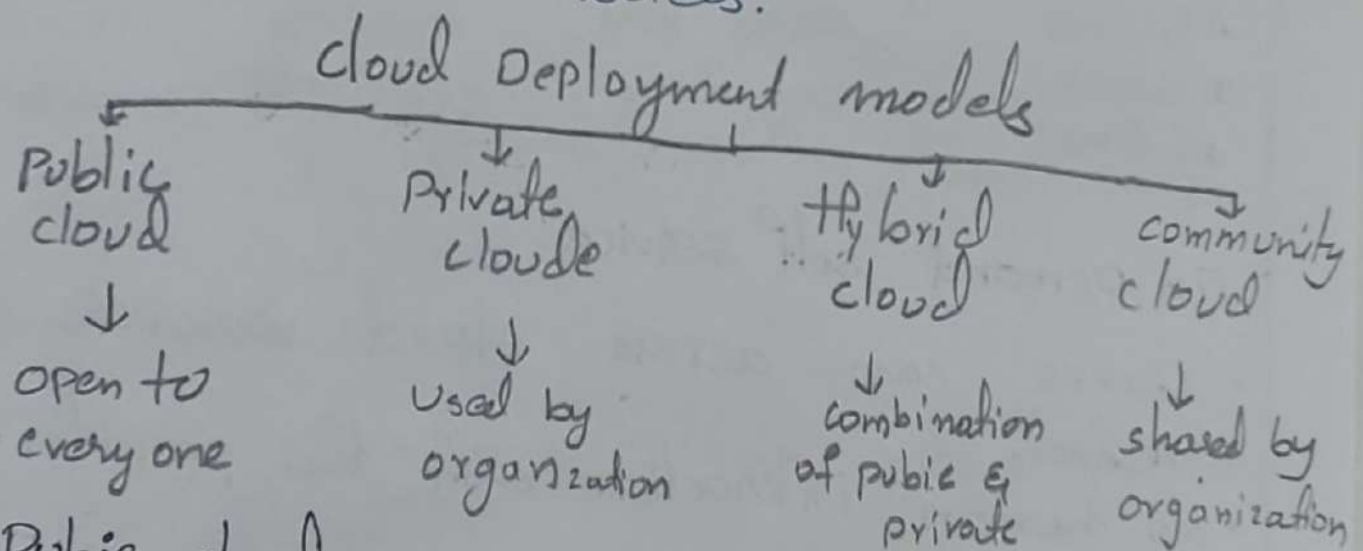
- Resources are shared among multiple users.
- Improves resource utilization and efficiency.

Rapid Elasticity:

- Resources can be increased or decreased quickly.
- Supports changing workload demands.

3 Deployment models: Map public cloud, private cloud, hybrid cloud and community cloud with usage scenarios.

cloud Deployment models:



Public cloud:

- Available for anyone over the internet.
- Suitable for websites and online applications.

Private cloud:
used by a single organization.
provides better security and control.

Hybrid cloud:

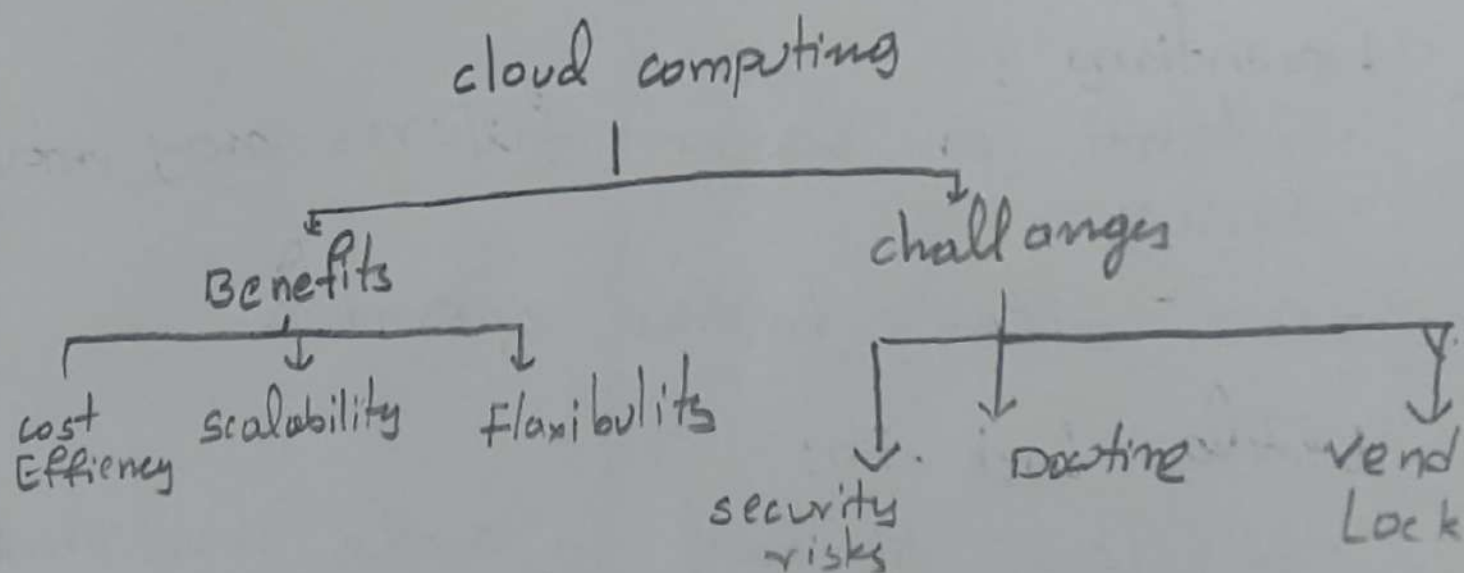
- combination of public and private clouds.
- Balances flexibility with security.

Community cloud:

- shared by organizations with similar requirements.

- Commonly used in healthcare and education

④ Cloud Benefits vs challenges: show advantages and limitations



Benefits:

cost efficiency:

- Reduces hardware and maintenance costs.
- pay only for the resources used.

Scalability:

- Resource can be expanded or reduced easily.
- supports business growth.

flexibility:

- services can be accessed from anywhere.
- Enables remote work and collaboration.

Challenges:

security:

- Risk of data breaches and cyberattacks.
- Requires strong security measures.

Downtime:

- Internet or server failures may interrupt services.
- Can affect business operations.

Vendor lock-in:

- Difficult to switch to another cloud provider.
- May involve extra cost and effort.